

Polynomials – 6

Polynomials over finite fields

Homework

8 сентября 2026 года

Theoretical minimum. The number of roots and $x^q - x = \prod (x - a)$; cyclicity of $\mathbb{F}_q^*$; the Frobenius automorphism; irreducible polynomials of degree d divide $x^{p^n} - x$ exactly when $d \mid n$; the vanishing derivative in characteristic p ; Chevalley's theorem on systems of low degree.

1. Prove the Chevalley–Warning theorem: if polynomials $f_1, \dots, f_r$ over $\mathbb{F}_p$ in n variables have $\sum \deg f_i < n$ and a common zero, then they have at least p common zeros. Deduce the Erdős–Ginzburg–Ziv theorem for a prime p .
2. (RMO) Let K be a finite field, $n > 2$, let $f \in K[X]$ be irreducible of degree n , and let g be the product of all non-constant polynomials in $K[X]$ of degree at most $n - 1$. Prove that f divides $g - 1$.
3. (RMO) Let K have $q = p^n$ elements and for $a \in K$ put $f_a = X^q - X + a$. Show that
 - a) $(X^q - X)^q - (X^q - X)$ is divisible by f_a ;
 - b) f_a has at least p^{n-1} essentially different irreducible factors in $K[X]$.
4. (RMO 2006) Let K be a finite field. Prove that the following are equivalent: $1 + 1 = 0$; and for every $f \in K[X]$ with $\deg f > 1$ the polynomial $f(X^2)$ is reducible.
5. (RMO) Let K be a finite field with $q > 3$ elements and let M be the set of polynomials in $K[X]$ of degree $q - 2$ whose coefficients are non-zero and pairwise distinct. Find the number of polynomials in M having $q - 2$ distinct roots in K .

6. (RMO) Let p be a prime, K a field of characteristic p with p^n elements, and $f_m = \sum_{k=0}^m (-1)^{m-k} \binom{m}{k} X^{p^k}$. Determine the set of roots of f_1 , and then the set of roots of f_m in K .
7. (S.T. Yau CSMC 2013) Find all irreducible polynomials of degree 2 and of degree 3 over $\mathbb{F}_2$, and determine the number of irreducible polynomials of degree 6 over $\mathbb{F}_2$.
8. (Iranian competition) Find the greatest common divisor of $4x^4 - 2x^2 + 1$ and $-3x^3 + 4x^2 + x + 1$ in $\mathbb{Z}_7[X]$.
9. (Сибирская 2022) Find the smallest order of a finite field F with the following property: there is a polynomial $f \in F[X]$ which is divisible by its own derivative and has exactly 2022 distinct roots in F .
10. (Putnam 2008) Let p be a prime and let $h(x) \in \mathbb{Z}[x]$ be such that $h(0), h(1), \dots, h(p^2-1)$ are distinct modulo p^2 . Show that $h(0), h(1), \dots, h(p^3-1)$ are distinct modulo p^3 .